

# Rev80 — Project Progress

Maps client meetings, stated requirements, and development phases from `VibeGui Project.md` to git commits. Useful for tracking what has been delivered and what remains.

## Project Background

Originated 2025-01-23 when a Digiducer-based data collection script was demonstrated to Carmen at a client site. The goal: replace expensive commercial vibration analysis tools (e.g. VibeAnalyze, \$999) with a purpose-built Python desktop app targeting industrial predictive maintenance workflows.

## Development Phases

### Proto — Initial prototype (Feb – Aug 2025)

Bare-bones data capture and display using the Digiducer USB audio interface. No HDF5, no velocity spectrum, minimal GUI.

| Date       | Commit  | Description                                       |
|------------|---------|---------------------------------------------------|
| 2025-02-15 | ae1736e | Project created                                   |
| 2025-03-05 | 5b017d1 | Project management notes                          |
| 2025-03-17 | 891d26f | Backup snapshot                                   |
| 2025-05-20 | cca1d4e | Working demo (Digiducer stream + basic plot)      |
| 2025-05-20 | e247dba | Module init                                       |
| 2025-05-21 | 856f855 | Digiducer abstracted into sensor and logger types |
| 2025-08-12 | 1257994 | First working dearpygui GUI with live stream      |
| 2025-08-12 | a3c2fd1 | Project management                                |
| 2025-08-12 | be9e91c | Bug fix: array dimensions                         |

### Meeting — 2025-01-23: Carmen, first demo

#### Requirements captured:

| #  | Requirement                                                      | Status            |
|----|------------------------------------------------------------------|-------------------|
| R1 | Frequency range and bin size selector                            | ✖ Done — Dec 2025 |
| R2 | Normalize FFT to IPS or g (velocity preferred over acceleration) | ✖ Done — Dec 2025 |
| R3 | Display top peaks                                                | Done — Dec 2025   |
| R4 | Smoothing                                                        | Not implemented   |
| R5 | Interactive peak inspection                                      | Not implemented   |
| R6 | Running speed harmonics markers                                  | Not implemented   |

| #   | Requirement                                                        | Status          |
|-----|--------------------------------------------------------------------|-----------------|
| R7  | Bearing fault frequency markers                                    | Not implemented |
| R8  | Metadata: date, company, machine info, bearing types, running rate | Not implemented |
| R9  | Trend: overall vibration over time                                 | Done — Mar 2026 |
| R10 | Luxury: startup and coastdown trend                                | Future          |

## Phase 0 — Digiducer acquisition, GUI polish (Nov – Dec 2025)

Core pipeline stabilised; units, HDF5 persistence, and modular architecture added.

| Date          | Commit            | Description                                                              |
|---------------|-------------------|--------------------------------------------------------------------------|
| 2025-11-17    | 587e393           | Save button added                                                        |
| 2025-11-20    | e9120d2           | Logging module with YAML config; print statements removed                |
| 2025-11-25    | c8bb0ae           | VibeSample refactored                                                    |
| 2025-11-26    | 6049f7f           | Carmen meeting notes — deployment roadmap                                |
| 2025-12-11    | 22014af           | Units switching; save/load via pickle                                    |
| 2025-12-16    | 10e48a0           | scipy FFT tools; GUI unit and integration controls                       |
| 2025-12-17    | 580d254           | Logger made universal                                                    |
| 2025-12-17    | 383f28e           | File browser fix                                                         |
| 2025-12-17    | 14dd5ca           | Unit conversion consistency                                              |
| 2025-12-17–18 | 772214e – 4424d3d | AcquisitionSettings upgrade (multi-step)                                 |
| 2025-12-29    | 38f22d3           | Plot units upgrade — units switching throughout GUI                      |
| 2025-12-29    | 8fb488c           | refactor_util — split into sample, sensor, simulation, digiducer modules |

## Meeting — 2025-11-25: Carmen, deployment roadmap

Phase definitions agreed:

| Phase    | Scope                                  | Budget |
|----------|----------------------------------------|--------|
| Phase 1  | Development to date (Digiducer era)    | \$2k   |
| Phase 2  | Polish and deploy to Windows           | \$1k   |
| Phase 2b | Integrate BNC DAQ for proximity probes | \$1k   |
| Phase 3  | Sales                                  | —      |

Requirements from this meeting:

| #   | Requirement                                                  | Status                                |
|-----|--------------------------------------------------------------|---------------------------------------|
| R11 | Switch between velocity and acceleration in frequency domain | Done — Dec 2025 ( 10e48a0 , 38f22d3 ) |

| #   | Requirement                        | Status                                      |
|-----|------------------------------------|---------------------------------------------|
| R12 | IPS zero-to-peak trend analysis    | Done — Jan 2026, extended Mar 2026          |
| R13 | RMS amplitude option               | Done — Mar 2026 ( 5872cab , 1e040d3 )       |
| R14 | Two simultaneous sensors (phasing) | Done — Mar 2026 (multi-channel pipeline)    |
| R15 | Proximity probe / DAQ support      | Partially done — PicoScope replaces BNC DAQ |

## Meeting — 2025-12-30: Carmen

### Requirements from this meeting:

| #   | Requirement                               | Status                                           |
|-----|-------------------------------------------|--------------------------------------------------|
| R16 | Overall vibration energy (IPS 0-P)        | Done — Jan 2026 ( ff4b31d )                      |
| R17 | Trend plot of overall vibration over time | Done — Mar 2026 ( cff702e )                      |
| R18 | Fix IPS unit scaling (was ~200× wrong)    | Done — Dec 2025 / confirmed Jan 2026 ( e7eddde ) |

## Phase 1b — HDF5, FFT validation, overall vibration (Jan 2026)

| Date       | Commit  | Description                                               |
|------------|---------|-----------------------------------------------------------|
| 2026-01-05 | dc64e37 | UI_Elements moved to util for global access               |
| 2026-01-05 | e3a988d | AcquisitionSettings hot-swap while streaming              |
| 2026-01-05 | cde0834 | Reorg and cleanup                                         |
| 2026-01-06 | 32ab828 | HDF5 save/load (replaces pickle)                          |
| 2026-01-09 | e76fb4f | DAQ hardware evaluation notes committed                   |
| 2026-01-10 | f0c549e | ISO timestamp; monotonic rel_time per block               |
| 2026-01-11 | e7eddde | FFT amplitude scaling validated with single-tone tests    |
| 2026-01-12 | ff4b31d | Overall 0-P vibration display (acceleration and velocity) |
| 2026-01-12 | fcf9e7e | Oversample stream for better Welch FFT resolution         |
| 2026-01-16 | 70c51db | uldaq dependency added (MCC USB-1608FS-Plus evaluation)   |
| 2026-01-20 | 2f95891 | MCC DAQ code examples                                     |
| 2026-03-09 | 2508756 | MCC DAQ integration work                                  |

## Meeting — 2026-01-11

**Notes:** - Overall 0-P vibration display added and demonstrated - Bin selector behaves poorly — flagged - Timestamp now FAT32-compatible (colons replaced by hyphens) - UI/UX language and jargon should match Alta's tooling - Demo screenshots needed for website

| #   | Requirement                                                         | Status                                                          |
|-----|---------------------------------------------------------------------|-----------------------------------------------------------------|
| R19 | Confirm overall vibration values are correct (vs. individual peaks) | <sup>*</sup> Done — validated <code>e7eddde</code>              |
| R20 | Fix bin selector behaviour                                          | Done — Mar 2026 ( <code>a69f14f</code> , <code>3fde4bc</code> ) |
| R21 | UI/UX language aligned with Alta                                    | Ongoing                                                         |
| R22 | Demo screenshots for website                                        | Not committed                                                   |

## Phase 2 — PicoScope 4000A acquisition backend (Mar 2026)

Hardware pivot: replace Digiducer/sounddevice with PicoScope 4000A. Enables any IEPE sensor (accelerometers, proximity probes, microphones) via standard BNC input — removes Digiducer hardware dependency.

| Date       | Commit                                    | Description                                                                                                                                                                                                              |
|------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-03-13 | <code>cf89848</code>                      | <b>PicoScope 4000A acquisition backend</b> — <code>FindPicoScope()</code> , <code>PicoScopeStream</code> , ADC → mV, accumulator, USB power fallback                                                                     |
| 2026-03-13 | <code>1b711c9</code>                      | Merge feature/picoscope into main                                                                                                                                                                                        |
| 2026-03-13 | <code>2b9aa38</code> <code>d9add87</code> | Comprehensive README added                                                                                                                                                                                               |
| 2026-03-13 | <code>91b06ce</code>                      | CLAUDE.md architecture guide added                                                                                                                                                                                       |
| 2026-03-16 | <code>8abb72d</code>                      | <b>IEPE sensor registry, triax support, PicoScope GUI integration</b> — <code>ScopeSensor</code> dataclass, <code>ScopeSensorRegistry</code> YAML persistence, per-channel sensitivity (mV/EU), GUI sensor library panel |
| 2026-03-16 | <code>cd40f63</code>                      | Harden PicoScope detection: overvoltage recovery, watchdog (5 s timeout, 3 reconnect attempts)                                                                                                                           |
| 2026-03-16 | <code>f03255d</code>                      | Fix handle-close between retries; extended delay after <code>PICO_NOT_RESPONDING</code>                                                                                                                                  |
| 2026-03-16 | <code>67ebfd6</code>                      | Test: update for new default voltage range                                                                                                                                                                               |

## Phase 2a — Multi-channel EU pipeline and GUI overhaul (Mar 2026)

Full multi-channel data path from hardware to display, with per-channel sensor assignment, modality-aware integration, and three-column GUI layout.

| Date       | Commit               | Description                                                                                           |
|------------|----------------------|-------------------------------------------------------------------------------------------------------|
| 2026-03-18 | <code>0d8911c</code> | Three-column GUI layout with primary window resizing                                                  |
| 2026-03-18 | <code>0147231</code> | Modality-aware integration (acc↔vel↔disp via FFT); mV/ EU sensitivity pipeline; unit conversion tests |

| Date       | Commit  | Description                                                                                                                                                                |
|------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-03-19 | 13b1095 | Add mil (1/1000 in) as displacement unit                                                                                                                                   |
| 2026-03-20 | afa5ce0 | Fix frequency-domain integration using signed exponent                                                                                                                     |
| 2026-03-24 | 09cea74 | <b>Multi-channel pipeline, scope sensor registry, EU unit refactor</b> — PicoScopeStream multi-channel buffers; per-channel mV → EU via ScopeSensor; DataCollector fan-out |
| 2026-03-24 | 4f8897c | <b>GUI overhaul</b> — channel config panel; per-channel status lights; dual-axis plots; scope sensor registry UI                                                           |
| 2026-03-24 | 8280bd5 | Test restructure: multi-channel, hardware isolation, new API coverage                                                                                                      |
| 2026-03-24 | 5d57ab4 | Fix PicoScope circular buffer wraparound in streaming callback                                                                                                             |
| 2026-03-24 | 0ca1251 | Replace Acquire/Configure tabs with streamlined left-panel layout                                                                                                          |
| 2026-03-24 | c70b8f3 | Visual section containers (borders, themed backgrounds) for left panel                                                                                                     |
| 2026-03-25 | cff702e | <b>Multi-channel data pipeline</b> — 32-frame ring cache; ChannelResult frozen dataclass; trend store; multi-channel HDF5 layout; browse ← → buttons                       |
| 2026-03-25 | 142dc30 | Expanded multi-channel test coverage; TestFrameCache, TestTrend, TestChannelResult                                                                                         |
| 2026-03-25 | 32410b6 | Fix: overflow bitmask; default voltage range 10 → 7 (10× finer ADC resolution)                                                                                             |
| 2026-03-25 | 23d0902 | Test: overflow key handling, per-channel helpers                                                                                                                           |
| 2026-03-25 | e365099 | Per-channel AC/DC coupling; results card sizing; 3 default peaks                                                                                                           |
| 2026-03-25 | 5ec3e45 | Fix: reset channel config and plot series on device switch                                                                                                                 |

## Phase 2b — Signal generator, spectrum controls, GUI polish (Mar 2026)

AWG output for excitation testing; FFT window selection; manual axis control; left-panel redesign; siggen config persistence.

| Date       | Commit  | Description                                                                                                                                                    |
|------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-03-25 | f74dd41 | <b>Signal generator UI</b> — Generate config tab (enable, waveform, freq, amplitude, offset); SimulatedSensor buried (CI/test only via VibeSensor.simulated()) |
| 2026-03-25 | 2a05ff9 | FFT window type selector (Hann/Blackman-Harris/Flatop/Hamming/Boxcar/Bartlett)                                                                                 |
| 2026-03-25 | 7e6b89f | Manual axis control — remove auto-fit on redraw; Autoscale button; Clear Cache button; full Welch frequency range (remove maxfreq display crop)                |
| 2026-03-25 | ffb4121 | Left panel rework — Device card, Channels card, button visibility fix (width=-1 bug), Load button restored                                                     |

| Date       | Commit  | Description                                                                                                                                                                                                                        |
|------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-03-25 | 83a980d | Card heights (DPG autosize_y fix); toggle button colour theming (idle/waiting/active); autoscale on first acquisition                                                                                                              |
| 2026-03-25 | ef0b560 | Channel Setup and Signal Generator buttons in Channels card                                                                                                                                                                        |
| 2026-03-25 | d0c1068 | Device card expanded with model, S/N, ID, channel count                                                                                                                                                                            |
| 2026-03-25 | 7bc729d | CH_WARNINGS_SECTION height scales to overflow channel count                                                                                                                                                                        |
| 2026-03-25 | eb811b6 | Siggen config persisted to <code>channel_assignments.yaml</code> ; restored on device connect                                                                                                                                      |
| 2026-03-25 | 7aa69b3 | <b>Unified VibeSample pipeline</b> — <code>process()</code> replaces <code>get_accel()</code> + <code>fft()</code> ; returns <code>ChannelResult</code> ; cross-modality FFT integration for physically correct time-domain signal |
| 2026-03-25 | 5872cab | Amplitude mode selector (RMS / 0-P / P-P) in GUI; Y-axis labels per mode                                                                                                                                                           |
| 2026-03-25 | 1e040d3 | Amplitude mode moved to <code>ScopeSensor</code> (per-sensor, not global spectrum setting)                                                                                                                                         |
| 2026-03-26 | a69f14f | <b>Modernised AcquisitionSettings</b> — derived properties (samplerate, blocksize from maxfreq/binsize); explicit highpass/lowpass filter fields; Welch overlap; <code>n_fft_bins</code> ; <code>memory_bytes</code>               |
| 2026-03-26 | ad98816 | Lowpass filter added to collector pipeline                                                                                                                                                                                         |
| 2026-03-26 | 3fde4bc | Spectrum tab redesign — combo presets; live-updating derived fields; filter controls                                                                                                                                               |
| 2026-03-26 | 36db391 | Tests updated for new AcquisitionSettings API                                                                                                                                                                                      |

## Meeting — 2026-03-18

| #   | Requirement                                         | Status                                                          |
|-----|-----------------------------------------------------|-----------------------------------------------------------------|
| R23 | Multi-channel: toggle channels on/off               | Done ( 09cea74 , 4f8897c )                                      |
| R24 | Two-channel phase relationship display              | Not implemented                                                 |
| R25 | Time-domain zoom window (e.g. 5 ms)                 | Not implemented                                                 |
| R26 | Track down velocity integration low-frequency noise | Partially addressed ( afa5ce0 ); warrants further investigation |
| R27 | Windows packaging (exe)                             | Not implemented                                                 |

## Meeting — 2026-03-25

| #   | Requirement                         | Status                     |
|-----|-------------------------------------|----------------------------|
| R28 | 0-P, P-P, RMS selectable per sensor | Done ( 5872cab , 1e040d3 ) |

| #   | Requirement                                     | Status                                                                                                                              |
|-----|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| R29 | Acquisition settings causing long capture times | Done — AcquisitionSettings now derives samplerate / blocksize from maxfreq / binsize with correct power-of-2 arithmetic ( a69f14f ) |
| R30 | Displaying wrong bins in spectrum setup         | Done — Spectrum tab redesigned with live derived fields ( 3fde4bc )                                                                 |

## Phase 4 — mV-domain pipeline + trend unit reflow (May 2026)

VibeSample refactored to store raw mV throughout; sensitivity conversion and Butterworth filtering moved to DataCollector.process\_sample. Welch PSD cached per sample (keyed to filter/welch config); five broadband RMS overalls (integration orders  $-2 \dots +2$ , in mV RMS) cached per sample and stored per trend timestep as a (M,5) ndarray. Unit/sensitivity/amplitude-mode changes reflow the trend at display time via DataCollector.get\_trend\_for\_display() — no clearing required. DataCollector.process\_samples() introduced as the single entry point for frame processing. GUI display loop simplified. HDF5 bumped to v4 with per-channel orders matrix; v3 files loaded in best-effort degraded mode.

| Date       | Commit  | Description                                                                   |
|------------|---------|-------------------------------------------------------------------------------|
| 2026-05-05 | 0c76c9a | refactor: mV-domain VibeSample + 5-order trend reflow                         |
| 2026-05-05 | 5fcf09a | refactor(gui): simplify display loop via process_samples()                    |
| 2026-05-05 | 3717286 | test: update for mV-domain pipeline                                           |
| 2026-05-05 | 0a1b38c | feat(gui): configurable frame cache depth with recording-window display       |
| 2026-05-27 | 90a97c6 | feat(picoscope): enumerate all connected scopes; hardware channel count probe |

## Requirements added in Phase 4

| #   | Source   | Requirement                                                                                                                              |
|-----|----------|------------------------------------------------------------------------------------------------------------------------------------------|
| R28 | May 2026 | Configurable frame cache depth (integer selector in Acquisition config tab) with live recording-window and total-memory derived displays |

## Requirements Tracker

| #  | Source   | Requirement                                                   | Status                                                                  |
|----|----------|---------------------------------------------------------------|-------------------------------------------------------------------------|
| R4 | Jan 2025 | Spectral smoothing                                            | Abandoned                                                               |
| R5 | Jan 2025 | Interactive peak inspection (click peak → identify frequency) | Done — crosshairs on spectrum, time-series, and trend plots ( 7f9b6a8 ) |



| #   | Source   | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Status                                      |
|-----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| R32 | Aug 2026 | <p>Anti-aliasing protection — field incident: high-frequency bearing fault aliased at low power, injecting spurious spectral energy. Fixed via mandatory oversample → zero-phase digital anti-alias filter → decimate in <code>PicoScopeStream</code>, plus a widened Nyquist margin (<code>samplerate = nextpow2(2.56*maxfreq)</code>, the 2.56x convention real FFT vibration analyzers use) so the filter has real transition-band room; retired the old user-facing (off-by-default, ineffective) lowpass control. Related fixes from the same hardware investigation: a new streaming-rate watchdog flags sustained USB throughput degradation that the existing silence watchdog couldn't see, and <code>MAXFREQ_PRESETS</code> ' top three entries (100k/250k/500k Hz) were dropped — they requested sample rates far beyond this hardware's measured continuous-streaming ceiling and were already silently corrupting most captured data.</p> | Done — see working tree (not yet committed) |
| R33 | Aug 2026 | Flexible plot layout / dockable panels — users need context-dependent layouts (e.g. spectrum + waveform vs. trend + spectrum). Investigate DPG drag-drop window docking; if not natively supported, evaluate panel-switching or split-view alternatives.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | TODO                                        |
| R34 | Aug 2026 | <p><b>ISO 20816-3 harmonization</b> — evaluation of machine vibration measured on non-rotating parts, for industrial machines &gt;15 kW at 120–15000 rpm: the class covering the blowers and motors this tool targets. Needs velocity RMS over a <i>declared</i> 10–1000 Hz band (2–1000 Hz below 600 rpm), machine-class selection, and Zone A/B/C/D boundaries with zone-based alarming. Current gaps: the overall is not band-limited — it spans <code>highpass_fc ... fs/2</code>, up to <math>2.05 \times F_{\text{max}}</math>, measured at +25% error from content the user explicitly excluded via <code>F_max</code> — and there is no machine classification or zone display, so the app reports a velocity number the user cannot interpret without an external table.</p>                                                                                                                                                                  | TODO                                        |

| #   | Source   | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Status |
|-----|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| R35 | Aug 2026 | <b>ISO 2954 instrument conformance</b> — requirements for instruments measuring vibration severity. Needs $\pm 10\%$ amplitude accuracy over 10–1000 Hz, a measurement band that is declared and stored with the data, frequency response within tolerance at the band edges, and overload indication. Current gaps: the band is undeclared and not persisted; the 4th-order highpass at 10 Hz puts $-3$ dB exactly on the lower band edge rather than in the passband (corner should sit at $\sim 2$ – $5$ Hz for a 10 Hz edge, or the response be compensated). Overload detection/exclusion is already done. | TODO   |
| R36 | Aug 2026 | <b>ISO 13373-1/-2 conformance</b> — condition monitoring: measurement procedures (-1) and processing/presentation of vibration data (-2). Substantially met already: the v4/v5 HDF5 stores the full acquisition, channel and sensor snapshot alongside the data. Remaining: a machine / measurement-point hierarchy and route concept (currently only free-text channel names and notes), and preventing or flagging a trend assembled from frames acquired under different $F_{\max}$ or band settings, which is not comparable.                                                                               | TODO   |
| R37 | Aug 2026 | <b>ISO 5348 mounting guidance</b> — mechanical mounting of accelerometers. Documentation rather than code: usable frequency range is dominated by the mount (stud $\gg$ adhesive $\gg$ magnet $\gg$ handheld probe), and a handheld probe is unusable much above $\sim 1$ kHz. The app will happily display a 10 kHz spectrum the mount cannot physically support, with no indication. Add mounting guidance to the README, and ideally a per-measurement mount-type field driving a usable-bandwidth warning on the spectrum.                                                                                  | TODO   |
| R38 | Aug 2026 | <b>Calibration traceability (ISO 16063)</b> — ScopeSensor carries a sensitivity typed in from the sensor datasheet, with no calibration date, method, or reference standard, so a saved measurement cannot be tied to a traceable calibration. Add calibration metadata to the sensor record and persist it with each measurement. A calibrated shaker check (0.1 in/s @ 191 Hz, 100 mV/g) put the end-to-end chain within $\sim 1.6\%$ , so the accuracy is there — the provenance is not.                                                                                                                     | TODO   |

| #   | Source   | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Status                              |
|-----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| R41 | Aug 2026 | <b>Envelope / demodulation analysis</b> — for rolling-element bearings this is <i>the</i> diagnostic: a defect's impulses ring a housing resonance at 2–20 kHz and are buried under the 1x in the raw spectrum, but appear as a clean line at the defect rate with $\pm 1x$ load-zone sidebands in the envelope, months earlier. Implemented in <code>rev80/envelope.py</code> with an Envelope plot tab and automatic demodulation-band selection. Verified against the simulated oracle at 125x SNR. Remaining: trending and alarming on the envelope line, which needs a defect-rate input (bearing geometry or a tachometer) and so depends on R6/R24; and drawing the demodulation band edges on the spectrum plot.                                                                                                                                                                                                                                                                                                                                                                                                                                     | Done (band trending pending R6/R24) |
| R42 | Aug 2026 | <b>Crest factor and kurtosis</b> — two impulsiveness scalars a broadband overall averages away entirely. Crest factor rises early in a bearing defect's life and falls once it spalls; kurtosis above ~4 flags repetitive impacts. On the result card, trended and persisted. Remaining: no trend <i>plot</i> (the trend plot's two y-axes are unit-based and these are dimensionless — needs a third axis, an R33 layout decision) and no alarming ( <code>FixedThresholdHook</code> is unit-aware and would need a dimensionless mode — belongs with R39).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Done (plot/alarm pending R33/R39)   |
| R39 | Aug 2026 | <b>Spectral anomaly hook — fix or remove.</b><br>Unwired from the GUI panel ( <code>GUI_ANOMALY_HOOK_TYPES</code> ); the hook and the headless front end are untouched. It triggers on <code>np.any(abs(spec - baseline)/baseline &gt; threshold)</code> across every bin, but Welch runs a single segment in every shipped preset ( <code>nperseg == blocksize</code> ), so each noise-floor bin is chi-squared(2) with a standard deviation equal to its own mean. P(some bin of ~2000 exceeds 1.5x) is ~1.0 on healthy data, which makes <code>consecutive_n</code> a delay rather than a defence; observed firing at the earliest frame it arithmetically can. Bins 0 and 1 are additionally hard-zeroed for integration, so on any velocity/displacement channel they deviate by ~1e12 and fire permanently. A real fix is band RMS rather than per-bin, a threshold in sigma rather than fixed %, persistence on the same band across frames, and a default band of <code>highpass_fc ... maxfreq</code> — i.e. it depends on R34's declared band. Decide then: rework on top of R34, or delete the hook and its GUI/headless/config surface outright. | TODO                                |

| #   | Source   | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Status |
|-----|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| R40 | Aug 2026 | <p><b>Sensor-fault detection.</b> Disconnect a cable mid-run today and the app trends a near-zero reading as a valid healthy measurement — the machine appears to have improved. The textbook fix, reading the IEPE bias against a nominal 8–14 V window, is <b>not available on this hardware</b>: our coupler has a DC blocking capacitor on its output so the bias never reaches the scope on either coupling setting, and the 4000A ranges only to <math>\pm 20</math> V, so a 24 V supply would over-range even with a direct pre-cap tap. Software workaround (in progress): dead-channel detection from the AC signal — flag a channel whose band RMS falls below a per- ScopeSensor minimum-plausible floor, plus the open-circuit signature of a rail-ward step decaying at the coupler's high-pass time constant. Threshold to be set from measured separation between connected/still, connected/excited, shorted and open, not invented. Hardware route, if ever wanted: a coupler tap ahead of the blocking cap with a divider bringing 24 V inside <math>\pm 20</math> V, or a coupler exposing a fault output.</p> | TODO   |

| #   | Source   | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| R43 | Aug 2026 | <p><b>Tachometer channel support</b> — shaft speed is the denominator that turns a spectrum into a diagnosis: it names the lines (a 5.43x bearing tone moves 6.1 Hz between no load and full load on a 4-pole motor, 12 bins at 0.5 Hz), separates 2x line frequency from 2x running speed (80 CPM apart, different repairs), and gates trending so a load swing is not read as a condition change. Scope for the first cut: RPM display and trend, order cursors on the spectrum, an Order column in the peaks table, and speed-gated alarming via <code>valid_results()</code>. <b>Deliberately out of scope:</b> order-normalised resampling (a smeared spectrum is rejected via <code>SPEED_DRIFT_MAX_PCT</code>, not corrected — bearing analysis is done at steady state) and balancing phase reference (the shipped high-pass rotates 1x by 21° at 3600 RPM rising to 143° at 600, while amplitude stays correct, so nothing on screen would flag the error). Measured accuracy limit: full accuracy needs <math>\geq \sim 70</math> samples per pulse (pulse rate <math>\leq \sim 600</math> Hz) — 36000 RPM at 1 ppr, 600 RPM at 60 ppr, 35 RPM at 1024 ppr, so a high-line encoder is impractical and the UI must say so. Storage is per-frame <b>edge times</b>, not the tach waveform — <math>\sim 1400\times</math> smaller and still enough to re-derive RPM at a different pulses/rev. Enables R6 and R24, and the envelope band trending left pending under R41.</p> | <p>Done — <code>rev80/tach.py</code>, channel roles, coherent simulation, collector wiring, HDF5 v5, speed gate, headless refusal, monitor v6, Tachometer tab with live preview, 1x marker and level. Closed out on a 4424A: 23 hardware tests pass, including an AWG sweep 300–10200 RPM within <math>\pm 0.2\%</math>, the AC-coupling failure reproduced electrically at 70% duty, and a clean 4-channel sustained run. Outstanding: shaft speed on the main display outside the dialog, and RPM on the trend plot (blocked on R33 — the trend already has two unit-based y-axes)</p> |
| R44 | Aug 2026 | <p><b>Tachometer support in <code>rev80-headless</code></b> — deferred, not rejected. Headless currently must <i>refuse</i> tach-role channels rather than ignore them: it builds its config from the same <code>devices/*.yaml</code> and iterates <code>enabled_channels</code>, so a GUI-configured tach channel would be high-passed, given an overall, trended and fed to the anomaly hooks as though it were vibration. Measured on a 5 % duty pulse train at 1800 RPM through the real <code>process_sample</code>: overall 1514.9 mV, crest 5.00, <b>kurtosis 15.94</b>, 63 spectral peaks — an analyst reviewing that session concludes a bearing is failing badly. It also drifts on nothing: a tach LED ageing from 5.0 V to 4.5 V moves that channel's overall by exactly <math>\sim 10\%</math>, the shipped <code>RmsThresholdHook</code> threshold, on three consecutive frames. Revisit once R43 lands.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | TODO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| #   | Source   | Requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Status                                                                    |
|-----|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| R45 | Sep 2026 | <b>Motor-state judgement from the tachometer</b><br>— a stopped shaft and a disconnected cable can produce an identical block: flat and near zero, whenever the reflector happens to sit away from the sensor. Only the parked-reflector case (flat at the high rail) and turning-below-the-floor (real pulses, fewer than MIN_EDGES) are distinguishable today, so <code>no_signal</code> is reported as exactly that and nothing is inferred about the machine. Making “motor stopped” trustworthy needs <i>history</i> — a channel that read cleanly and then ceased is a stop; one that never read is a setup problem — which is a small state machine belonging with whatever consumes the state rather than in <code>tach.py</code> . Until then the GUI must not claim a machine is stopped on the strength of a flat line.                                      | TODO                                                                      |
| R46 | Sep 2026 | <b>Surface velocity from reflector size and duty cycle</b> — with reflector arc length $L$ , measured duty $d$ and shaft rate $f$ , the reflector subtends $d$ of a revolution, so circumference is $L/d$ and surface velocity is $v = f \cdot L/d$ . The tape doubles as a shaft-diameter measurement, giving surface speed without anyone measuring the shaft — directly useful on rollers, belts and web handling. <b>Duty cycle is now captured</b> ( <code>TachResult.duty_cycle / .pulse_widths_s</code> , persisted in v5), so the remaining work is the reflector-size input and the $v = f \cdot L/\text{duty}$ readout. Sensor geometry — an optical tach's spot width, a proximity probe's inductive field — inflates the observed duty; by decision the operator accounts for that when measuring the tape or key, so no correction is applied in software. | TODO                                                                      |
| —   | Future   | Proximity probe support                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Done — scope sensor with EU in displacement units                         |
| —   | Future   | Web portal for data sharing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Abandoned                                                                 |
| —   | Future   | MCC DAQ tooling (USB-1608FS-Plus)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Abandoned — PicoScope oscilloscope replaces DAQ for all current use cases |

## Architecture Evolution Summary

| Era                                                | Hardware               | Key modules                                                                                                                                                                     | Format                                                                               |
|----------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Proto<br>(Feb–Aug 2025)                            | Digiducer USB audio    | monolithic                                                                                                                                                                      | —                                                                                    |
| Phase 0<br>(Nov–Dec 2025)                          | Digiducer              | sensor, sample, simulation, digiducer                                                                                                                                           | pickle                                                                               |
| Phase 1b<br>(Jan 2026)                             | Digiducer              | + HDF5, refactored AcquisitionSettings                                                                                                                                          | HDF5                                                                                 |
| Phase 2 (Mar 2026)                                 | <b>PicoScope 4000A</b> | + picoscope, scope_sensor, scope_sensor_registry                                                                                                                                | HDF5 multi-channel                                                                   |
| Phase 2a<br>(Mar 2026)                             | PicoScope              | + ChannelResult, multi-channel pipeline, trend                                                                                                                                  | HDF5 multi-channel                                                                   |
| Phase 2b<br>(Mar 2026)                             | PicoScope + AWG        | + signal generator, Welch controls, derived AcquisitionSettings                                                                                                                 | HDF5 + YAML config                                                                   |
| Windows packaging<br>(Mar–May 2026)                | PicoScope              | + <code>_paths.py</code> , <code>_pico_loader.py</code> , <code>build/collect_pico_dlls.py</code> ; PyInstaller + Inno Setup pipeline                                           | HDF5 v4 + frozen one-dir bundle                                                      |
| Phase 4 (May 2026)                                 | PicoScope              | mV-domain VibeSample, (M,5) trend, process_samples()                                                                                                                            | HDF5 v4                                                                              |
| Monitor Mode<br>(May–Jun 2026)                     | PicoScope              | + <code>monitor/ ( gate , controller , writer , anomaly , session )</code> , <code>headless.py</code> ; config split into <code>acquisition.yaml</code> + <code>devices/</code> | HDF5 v4 + one <code>session.h5</code> v5 per session                                 |
| Rebrand → Rev80 (Aug 2026)                         | PicoScope              | src-layout move to <code>src/rev80/</code> ; <code>doc/ / scripts/ / build/ reorg</code>                                                                                        | HDF5 v4; data <code>~/Documents/Rev80/</code> , config <code>~/.config/rev80/</code> |
| Anti-alias<br>(Aug 2026)                           | PicoScope 4424A        | <code>picoscope.antialias_decimate()</code> — mandatory oversample → Kaiser FIR → decimate; streaming-rate degradation watchdog; user-facing lowpass removed                    | HDF5 v4 + per-frame degraded flag                                                    |
| Audit round 1 — measurement validity (Aug 2026)    | PicoScope              | + <code>_dsp.py</code> ; persistent per-channel high-pass state, Hann/Tukey-tapered integration, <code>nperseg = blocksize</code> , true (float) sample rate                    | HDF5 v4, overflow/degraded read back                                                 |
| Audit round 2 — integrity & diagnostics (Aug 2026) | PicoScope              | + <code>envelope.py</code> , <code>peaks.py</code> ; declared measurement band, crest factor / kurtosis, <code>GenerateBearingVibration()</code> oracle                         | HDF5 v4 + band, crest/kurtosis trends, <code>peak_threshold_db</code>                |
| Spectral averaging (Aug 2026)                      | PicoScope              | <code>_psd_and_overalls_for()</code> ; power-domain averaging over N frames, above the per-frame PSD cache                                                                      | HDF5 v4 unchanged — the average is a view on stored frames                           |

| Era                                 | Hardware                 | Key modules                                                                                                                                                                                                                                                                                                             | Format                                                                                                                                |
|-------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| CLI & Linux packaging (Aug 2026)    | PicoScope                | + <code>desktop.py</code> (XDG launcher entry + hicolor icons); unified <code>rev80</code> CLI with a <code>headless</code> subcommand; <code>CONTRIBUTING.md</code> split out; <code>resource_path()</code> fixed for non-editable installs                                                                            | HDF5 v4; <code>data_dir()</code> always <code>~/Documents/Rev80/data</code> , in dev and frozen builds alike                          |
| Raw-stream retention (Aug–Sep 2026) | PicoScope                | fixed <code>RAW_SAMPLERATE_HZ</code> acquisition rate vs. derived display rate (exactly $2.56 \times \text{maxfreq}$ from Sep); <code>decimate_to_rate()</code> , <code>eu_scaled_raw()</code> , <code>current_frame()</code> , <code>simulation._RawRateView</code> , <code>scripts/validate-streaming-capacity</code> | HDF5 — every stored frame is the raw-rate capture, independent of <code>maxfreq</code>                                                |
| Tachometer (Sep 2026)               | PicoScope + optical tach | + <code>tach.py</code> , channel roles, speed gate, Tachometer tab, 1× marker, <code>GenerateTachPulse /</code> <code>GenerateMachineWithTach</code>                                                                                                                                                                    | HDF5 <b>v5</b> — edge times, no <code>data</code> on tach channels; <code>session.h5</code> <b>v6</b> ( <code>rpm / speed_ok</code> ) |